

WASP-43b

R-2685

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_130324 · created 2026-08-31T00:30:44+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456375.69553 +/- 0.00055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.057
Transit depth (Rp/Rs)^2	3.9 +/- 2.26 [%]
Orbital Inclination (inc)	86.8 +/- 2.93
Airmass coefficient 1 (a1)	1.55 +/- 0.79
Airmass coefficient 2 (a2)	-0.37 +/- 0.79
Scatter in the residuals of the lightcurve fit is	55.76 %
Best Comparison Star	None
Optimal Aperture	1.82
Optimal Annulus	3.0
Transit Duration (day)	0.0614 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.057 vs 0.1594 (deviation 0.66σ)
Duration fit vs published	0.0614 d vs 0.0512 d (deviation 2.15σ)
Mid-time O–C	0.4 min
Depth SNR	3.4
Residual scatter	55.76 %

Light curve

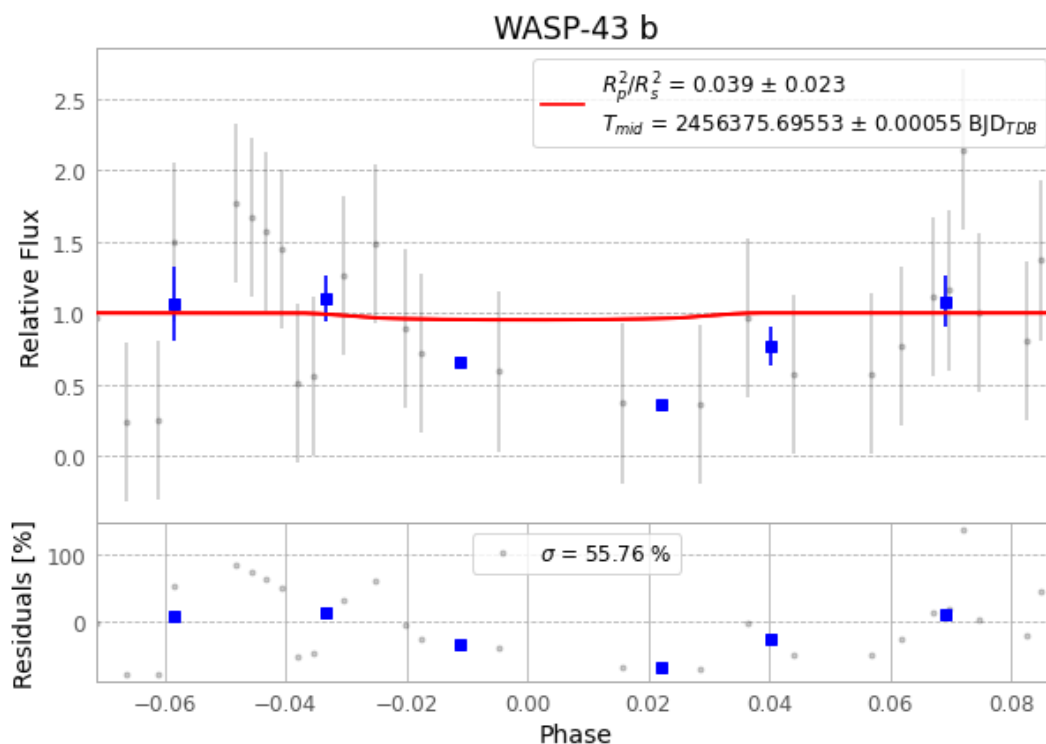


Figure 1 — FinalLightCurve_WASP-43 b_2013-03-23.png (SHA-256 c9346b074c4b2ef0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [460, 206], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 56984f80f6439262...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), WASP-43130324030912.FITS → WASP-43130324061512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-130324.log (f4df153dc84b...), FinalLightCurve_WASP-43 b_2013-03-23.png (c9346b074c4b...), FinalLightCurve_WASP-43 b_2013-03-23.csv (6dc1c7bedb16...)

Compute resources

Wall time 200.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-31 00:30Z.